

Online Judge Platform

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REST API Documentation Specification

Online Judge Platform — OpenAPI 3.0 Specification

| Document Metadata | Details |

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| **Project Title** | Online Judge Platform |

| **API Version** | v1.0.0 |

| **Base URL** | <https://api.onlinejudge.com/api/v1> |

| **Authentication Scheme** | Bearer Token (JSON Web Token) |

| **Format** | JSON (`application/json`) |

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1. Global Standard Responses & Headers

Request Headers

`Content-Type: application/json`

`Authorization: Bearer`

Standard Error Response Format

```
{
  "success": false,
  "error": {
    "code": "RESOURCE_NOT_FOUND",
    "message": "The requested problem with ID 64b8f... does not exist.",
    "timestamp": "2026-07-26T20:00:00.000Z"
  }
}
```

2. Authentication Endpoints

2.1 Register New User

Endpoint: `POST /auth/signup`

Access: Public

Description: Registers a new regular user account.

Request Body

```
{
  "name": "Jane Doe",
  "email": "jane.doe@example.com",
  "password": "SecurePassword123!"
}
```

Response (201 Created)

```
{
  "success": true,
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "user": {
      "id": "64b8f9e210a1b2001c8e4321",
      "name": "Jane Doe",
      "email": "jane.doe@example.com",
      "role": "User"
    }
  }
}
```

2.2 User Login

Endpoint: `POST /auth/login`

Access: Public

Description: Authenticates user credentials and returns a JWT access token.

Request Body

```
{
  "email": "jane.doe@example.com",
  "password": "SecurePassword123!"
}
```

Response (200 OK)

```
{
  "success": true,
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiI2NGI4ZjllMjEwYTFiMjAwMWM4ZTQzMjEiLCJyb2x1IjoiaVXNlciIsIm1hdCI6MTcwMDAwMDAwMCwiZXhwIjoxNzAwMDg2NDAwfQ...",
    "user": {
      "id": "64b8f9e210a1b2001c8e4321",
      "name": "Jane Doe",
      "email": "jane.doe@example.com",
      "role": "User"
    }
  }
}
```

- `limit` (default: 20)
- `difficulty` (`Easy` | `Medium` | `Hard`)
- `search` (keyword string)

Response (200 OK)

```
{
  "success": true,
  "meta": {
    "total": 150,
    "page": 1,
    "limit": 20,
    "totalPages": 8
  },
  "data": [
    {
      "id": "64b8f9e210a1b2001c8e9999",
      "title": "Two Sum",
      "slug": "two-sum",
      "difficulty": "Easy",
      "constraints": {
        "timeLimitMs": 1000,
        "memoryLimitMb": 256
      }
    }
  ]
}
```

3.2 Get Problem Details

Endpoint: `GET /problems/:id`

Access: Public

Response (200 OK)

```
{
  "success": true,
  "data": {
    "id": "64b8f9e210a1b2001c8e9999",
    "title": "Two Sum",
    "slug": "two-sum",
    "statement": "Given an array of integers nums and an integer target, return indices of the two numbers such that they add up to target.",
    "difficulty": "Easy",
    "constraints": {
      "timeLimitMs": 1000,
      "memoryLimitMb": 256
    },
    "sampleInput": "[2, 7, 11, 15]\n9",
    "sampleOutput": "[0, 1]"
  }
}
```

3.3 Create New Problem (Admin Only)

Endpoint: `POST /problems`

Access: Authenticated (`Admin`)

Request Body

```
{
  "title": "Reverse Linked List",
  "statement": "Given the head of a singly linked list, reverse the list, and return the reversed list.",
  "difficulty": "Easy",
  "constraints": {
    "timeLimitMs": 1000,
    "memoryLimitMb": 256
  },
  "sampleInput": "[1, 2, 3, 4, 5]",
  "sampleOutput": "[5, 4, 3, 2, 1]"
}
```

Response (201 Created)

```
{
  "success": true,
  "data": {
    "id": "64b8f9e210a1b2001c8e8888",
    "title": "Reverse Linked List",
    "slug": "reverse-linked-list",
    "createdBy": "64b8f9e210a1b2001c8e4321",
    "createdAt": "2026-07-26T20:30:00.000Z"
  }
}
```

4. Test Case Management Endpoints (Admin Only)

4.1 Create Test Case for Problem

Endpoint: `POST /testcases`

Access: Authenticated (`Admin`)

Request Body

```
{
  "problemId": "64b8f9e210a1b2001c8e9999",
  "input": "4\n2 7 11 15\n9",
  "expectedOutput": "0 1",
  "isHidden": true
}
```

Response (201 Created)

```
{
  "success": true,
  "data": {
    "id": "64b8f9e210a1b2001c8e7777",
    "problemId": "64b8f9e210a1b2001c8e9999",
    "isHidden": true
  }
}
```

5. Submissions Endpoints

5.1 Submit Code Solution

Endpoint: `POST /submissions`

Access: Authenticated ([User](#) / [Admin](#))

Request Body

```
{
  "problemId": "64b8f9e210a1b2001c8e9999",
  "language": "cpp",
  "sourceCode": "#include <iostream>\nusing namespace std;\nint main() {\n    int n; if(!(cin >>\n    n)) return 0;\n    int a[100]; for(int i=0;i<n;i++) cin >> a[i];\n    int target; cin >> target;\n    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]+a[j]==target){\n                cout << i << \" \" << j << endl;\n                return 0;\n            }\n        }\n    }\n    return 0;\n}"
}
```

Response (202 Accepted for Processing)

```
{
  "success": true,
  "data": {
    "submissionId": "64b8f9e210a1b2001c8e5555",
    "status": "Pending",
    "message": "Submission enqueued successfully for evaluation."
  }
}
```

5.2 Poll Submission Status & Verdict

Endpoint: [GET /submissions/:id](#)

Access: Authenticated ([User](#) / [Admin](#))

Response (200 OK - Processing Complete)

```
{
  "success": true,
  "data": {
    "id": "64b8f9e210a1b2001c8e5555",
    "problemId": "64b8f9e210a1b2001c8e9999",
    "language": "cpp",
    "verdict": "Accepted",
    "executionTimeMs": 14,
    "memoryUsedKb": 3480,
    "submittedAt": "2026-07-26T20:40:00.000Z"
  }
}
```

5.3 Get User Submission History

Endpoint: `GET /submissions/history`

Access: Authenticated (`User` / `Admin`)

Response (200 OK)

```
{
  "success": true,
  "data": [
    {
      "id": "64b8f9e210a1b2001c8e5555",
      "problemTitle": "Two Sum",
      "language": "cpp",
      "verdict": "Accepted",
      "executionTimeMs": 14,
      "submittedAt": "2026-07-26T20:40:00.000Z"
    }
  ]
}
```

6. Leaderboard Endpoints

6.1 Get Global Leaderboard

Endpoint: `GET /leaderboard`

Access: Public

Response (200 OK)


```
{
  "success": true,
  "data": [
    {
      "rank": 1,
      "userName": "Alex Mercer",
      "problemsSolved": 128,
      "totalSubmissions": 142,
      "accuracy": 90.14
    },
    {
      "rank": 2,
      "userName": "Jane Doe",
      "problemsSolved": 42,
      "totalSubmissions": 98,
      "accuracy": 42.85
    }
  ]
}
```